# The Hierarchical Oscillatory Information Framework: A Testable Dynamical Model of Biological State Drift and Structured Oscillatory Input

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**Abstract**

This paper proposes the Hierarchical Oscillatory Information Framework (HOIF), a computational and mathematical framework for testing whether structured oscillatory inputs can act as information-bearing variables in biological state dynamics. HOIF represents biological state as a multiscale vector $S(t)$, health as an attractor set $\mathcal{H}$, disease drift as $D(t) = \mathrm{dist}(S(t), \mathcal{H})$, and structured input as $A(t)$. The central claim is not that sound treats disease, but that waveform organization may be experimentally testable as a predictor of biological trajectories after energy, duration, heating, and nonspecific stress are controlled. The current disease-engine simulation remains illustrative because its drift parameters are assigned rather than inferred from data. Therefore, this manuscript explicitly separates the proposed framework from its toy implementation and outlines a data-driven version in which drift parameters are inferred from transcriptomics, chromatin accessibility, proteomics, cell mechanics, mitochondrial function, imaging, and single-cell measurements. The paper also defines a biologically interpretable recovery metric, uncertainty quantification, sensitivity-analysis requirements, competing baseline models, falsification criteria, and a re-entrainment theorem for a simplified contractive system. HOIF is presented as a falsifiable research program: it should be retained only if waveform structure improves prediction of biological state beyond energy-matched controls and if inferred drift parameters generalize to independent datasets.

**Keywords:** acoustic biology; cellular resonance; sonocytology; mechanotransduction; disease modeling; Kuramoto model; Peyrard–Bishop model; computational medicine

## 1 Introduction

Modern medicine already uses several channels to influence cellular behavior: chemical signaling through drugs, genetic intervention through gene editing or gene therapy, electrical stimulation through implanted or external devices, and optical control in specialized research systems. This paper asks whether a related but less developed channel—structured oscillatory input—is sufficiently plausible and mathematically coherent to justify systematic testing. The motivating idea is not that an acoustic code has already been found, but that biological systems are dynamic enough for such a possibility to be experimentally examined.

The hypothesis begins from a simple analogy. Human hearing is limited to a band of acoustic frequencies, yet other organisms detect ranges beyond human perception. Cells do not "hear" in the cognitive sense, but they are mechanical, electrical, and molecular systems capable of responding to vibration, pressure, stiffness, deformation, and oscillatory forcing. If cellular structures possess characteristic resonant modes, then some disease processes may involve altered dynamic signatures in addition to biochemical changes. A testable research program can therefore ask whether carefully controlled oscillatory inputs measurably shift diseased model systems toward healthier state descriptors.

This idea must be stated carefully. The simulations in this paper are not medical evidence, do not show that sound cures disease, and should not be interpreted as a substitute for laboratory or clinical testing. They are mathematical models designed to ask a narrower question: if disease is represented as layer-specific frequency drift, and if a restoring protocol is represented as frequency entrainment, what patterns emerge across many disease classes?

The present model does not establish treatment efficacy. Its value is that it generates a hierarchy of predictions. In the disease engine described here, metabolic, autoimmune, and some aging-associated conditions show higher modeled recovery, while cancer and genetic disorders show lower modeled recovery. This pattern suggests a prioritization strategy for experiments: begin where reversible dysregulation is plausible, and treat mutation-driven or malignancy-driven disease as a stringent boundary case rather than an early proof target.

Core Hypothesis: Hierarchical Oscillatory Information Framework (HOIF)

**Definition of the framework.** The Hierarchical Oscillatory Information Framework (HOIF) is the hypothesis that biological state can be modeled as a hierarchy of coupled dynamical systems in which structured mechanical or oscillatory inputs may act as one experimentally testable information channel.

**C1. Biological state.** A living system can be represented by a multiscale state vector $S(t)$ whose components describe organism, organ, tissue, cellular, organelle, nuclear, chromatin, and molecular variables.

**C2. Healthy attractor.** Health is represented not as one fixed point, but as a stable attractor set $\mathcal{H}$ in biological state space.

**C3. Disease drift.** Disease is represented as deviation $D(t) = \mathrm{dist}(S(t), \mathcal{H})$ from that attractor set.

**C4. Oscillatory input.** Structured oscillatory stimulation is represented as an input $A(t)$ with amplitude, frequency, phase, waveform, and temporal organization.

**C5. State dynamics.** The central mathematical claim is that biological trajectories may be modeled as $\frac{dS}{dt} = F(S, A, t) + \eta(t)$, where $\eta(t)$ captures biological noise.

**C6. Information hypothesis.** HOIF is not primarily a theory of sound; it is a theory that waveform structure may carry biologically relevant information after total energy, duration, and heating are controlled.

**C7. Unique prediction.** If two signals have identical energy, average frequency, and duration but different temporal organization, HOIF predicts that at least some biological systems may show distinguishable downstream state trajectories.

**C8. Research posture.** HOIF is not proposed as established biology or clinical therapy. It is proposed as a precise, falsifiable framework worth testing.

## 2 Relationship to Existing Literature

This framework sits at the intersection of several established fields while making a narrower and more speculative theoretical proposal.

**Established knowledge: mechanobiology and mechanotransduction.** Cells sense and respond to mechanical properties of their environment, including stiffness, geometry, force, and deformation. These cues can alter cytoskeletal organization, nuclear shape, signaling pathways, and gene expression [1, 2]. The present paper does not claim to discover mechanotransduction. It uses mechanotransduction as the biological basis for asking whether structured oscillatory inputs can be formalized as state-control signals.

**Established knowledge: cellular nanomechanics and ultrasound bioeffects.** Living cells exhibit measurable nanomechanical motion, and ultrasound can alter biological systems through mechanisms including radiation force, cavitation, membrane permeabilization, and thermal effects [3–6]. The present framework differs by focusing on waveform-dependent information content rather than treating acoustic exposure only as energy deposition.

**Established knowledge: nuclear mechanics and gene regulation.** The nucleus is mechanically integrated with the cytoskeleton, and nuclear deformation can influence chromatin organization and transcriptional behavior [7, 8]. This supports the plausibility of cross-scale mechanical coupling, but it does not prove that a therapeutic acoustic code exists.

**Established knowledge: systems biology, information theory, and oscillator models.** Biological regulation is often modeled as a networked dynamical system, and information-theoretic approaches have been used to analyze signaling specificity, noise, and channel capacity [9, 10]. Coupled-oscillator theory provides a mathematical language for synchronization and entrainment [13, 14]. The novelty here is to combine these ideas into a disease-state model where structured oscillatory input is treated as a candidate control signal.

**This paper's divergence.** Existing work generally asks how cells respond to mechanical force, ultrasound energy, or biochemical signaling. This paper asks a more specific question: can waveform structure itself be modeled as an information-bearing variable that changes biological state trajectories after energy, heating, and nonspecific stress are controlled?

## 3 Core Hypothesis

The proposed framework rests on four linked assumptions that are intended to be tested, revised, or rejected:

- **H1. Cellular structures have dynamic signatures.** Membranes, cytoskeletal networks, mitochondria, nuclei, and DNA-associated structures can be modeled as oscillatory systems with characteristic healthy dynamics.
- **H2. Disease includes frequency drift.** A diseased cell state may be represented as a shift away from healthy dynamics across one or more biological layers.
- **H3. External signals may entrain biological dynamics.** A controlled signal may, in some model systems, shift a diseased oscillator toward a target healthy frequency descriptor.
- **H4. Disease classes may differ in reversibility.** Conditions dominated by regulatory drift are predicted to be more testable first targets than conditions dominated by fixed genetic lesions or malignant cellular autonomy.

The phrase "acoustic protocol" is used broadly in this paper to describe externally applied oscillatory input. Depending on the frequency range and biological target, such input may correspond to audible sound, ultrasound, electromagnetic terahertz stimulation, or another physical delivery mechanism. The models here do not solve the delivery problem. They model the response problem.

## 4 Formal Definitions of HOIF

The central purpose of HOIF is to translate the verbal hypothesis into a compact mathematical object that can be tested or rejected.

**Definition 1: Biological state.** Let

$$S(t) = \big(S_O(t), S_{org}(t), S_T(t), S_C(t), S_{mit}(t), S_N(t), S_{chr}(t), S_{DNA}(t)\big) \tag{1}$$

be a hierarchical biological state vector, where the components represent organism-level, organ-level, tissue-level, cellular, mitochondrial, nuclear, chromatin, and DNA-associated variables. HOIF does not require every component to be measured in every experiment; it requires that the measured subset be explicitly defined.

**Definition 2: Healthy attractor.** Let $\mathcal{H} \subset \mathcal{S}$ be the set of states judged healthy for the experimental system under defined conditions. $\mathcal{H}$ is an attractor region, not a single idealized value.

**Definition 3: Disease drift.** Disease drift is the distance from the observed state to the healthy attractor:

$$D(t) = \mathrm{dist}(S(t), \mathcal{H}). \tag{2}$$

A model supports the framework only if $D(t)$ can be operationalized using measurable variables such as morphology, viability, inflammatory markers, mitochondrial function, chromatin state, or transcriptional profiles.

**Definition 4: Oscillatory input.** The structured input is

$$A(t) = \alpha(t) g\big(f(t), \phi(t), w(t), \tau(t)\big), \tag{3}$$

where $\alpha(t)$ is amplitude, $f(t)$ is instantaneous frequency, $\phi(t)$ is phase, $w(t)$ is waveform shape, and $\tau(t)$ denotes temporal ordering or motif structure.

**Definition 5: Dynamical law.** The general state equation is

$$\frac{dS}{dt} = F(S, A, t) + \eta(t), \tag{4}$$

where $F$ is the biological response operator and $\eta(t)$ is stochastic biological variability. HOIF becomes experimentally meaningful only when $F$ is estimated from controlled data.

**Definition 6: Re-entrainment criterion.** A candidate re-entrainment event is observed when

$$D(T) < D(0) \tag{5}$$

under structured input and when this reduction is larger than sham, random-waveform, and energy-matched controls.

**Definition 7: Information-specific effect.** A result supports the information component of HOIF only if waveform organization predicts biological outcome after controlling for energy, duration, average frequency, temperature, and nonspecific stress.

## 5 Conceptual Diagrams

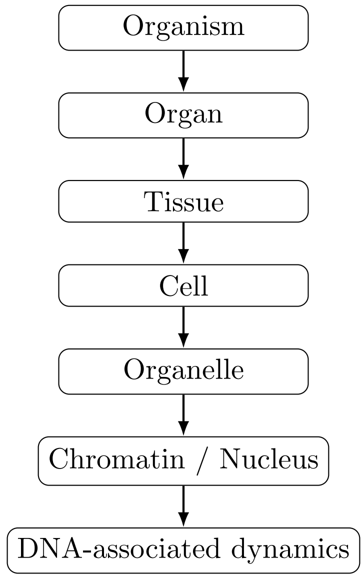



Figure 1: Biological hierarchy represented in HOIF. The framework treats biological state as multiscale rather than locating response at a single molecular layer.

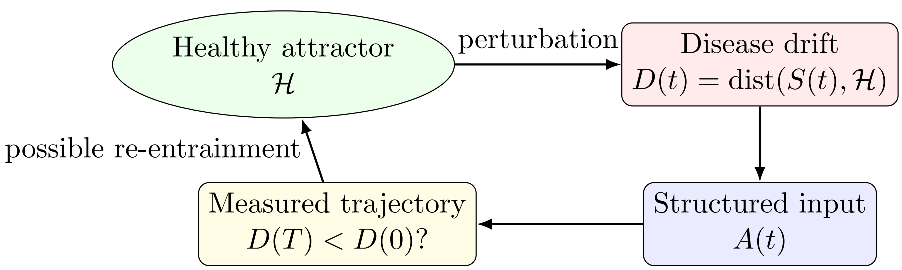



Figure 2: Healthy attractor, disease drift, structured oscillatory input, and the re-entrainment criterion. The diagram describes a testable modeling loop, not demonstrated therapy.

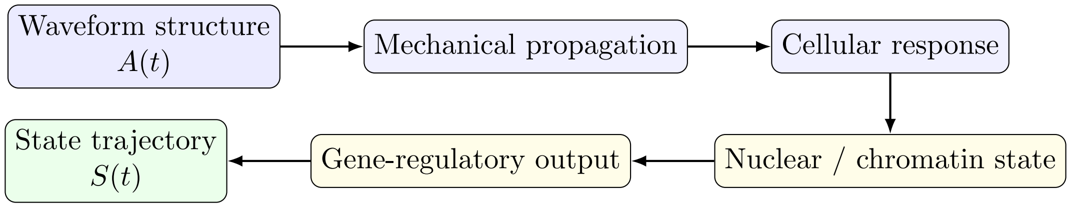



Figure 3: Information-flow hypothesis. HOIF asks whether waveform structure explains biological response beyond energy, heating, and nonspecific stress.

# 6 General Mathematical Framework

The disease engine can be generalized beyond the particular numerical simulation by defining biological state as a hierarchy of coupled state spaces. Let

$$x(t) = \big(x_m(t), x_c(t), x_{mit}(t), x_n(t), x_d(t)\big) \in \mathcal{X}, \tag{6}$$

where $x_m$, $x_c$, $x_{mit}$, $x_n$, and $x_d$ denote membrane, cytoskeletal, mitochondrial, nuclear, and DNA-associated state variables. The healthy state is not a single point but a bounded attractor region $\mathcal{A}_H \subset \mathcal{X}$. A disease state is represented as displacement from this region:

$$\Delta_D(t) = \mathrm{dist}\big(x(t), \mathcal{A}_H\big). \tag{7}$$

**Coupling operators.** Cross-scale biological influence is represented by coupling operators $C_{ij} : \mathcal{X}_j \to \mathcal{X}_i$, so that each layer evolves according to

$$\frac{dx_i}{dt} = F_i(x_i) + \sum_{j \neq i} C_{ij}(x_j, x_i) + B_i u(t) + \eta_i(t), \tag{8}$$

where $F_i$ is the intrinsic layer dynamics, $u(t)$ is the applied oscillatory input, $B_i$ maps the input into the relevant biological layer, and $\eta_i(t)$ represents biological noise.

**Information-bearing input.** The applied signal is not defined only by intensity. It is represented as

$$u(t) = A(t)\sin\big(\phi(t)\big), \qquad \frac{d\phi}{dt} = 2\pi f(t), \tag{9}$$

where amplitude $A(t)$, instantaneous frequency $f(t)$, phase $\phi(t)$, sweep geometry, and temporal ordering may all contribute to biological effect. The central theoretical claim is that two inputs with matched total energy,

$$\int_0^T u_1^2(t)dt = \int_0^T u_2^2(t)dt, \tag{10}$$

may produce different state trajectories if their waveform structures differ.

**Stability and convergence.** For modeling purposes, a candidate entrainment response is defined as a reduction in distance to the healthy attractor:

$$\Delta_D(T) < \Delta_D(0), \tag{11}$$

with stronger convergence when $\Delta_D(t)$ decreases monotonically or when the final state remains within a tolerance radius $\epsilon$ of $\mathcal{A}_H$:

$$\text{dist}\big(x(T), \mathcal{A}_H\big) < \epsilon. \tag{12}$$

For a local stability analysis, the controlled healthy region is stable if a Lyapunov-like function $V(x) = \text{dist}(x, \mathcal{A}_H)^2$ satisfies

$$\frac{dV}{dt} < 0 \tag{13}$$

for perturbed states in a neighborhood of the disease trajectory under the selected input.

**Information flow across scales.** In an experimental version of the framework, information flow could be estimated by asking whether the signal history $u_{0:t}$ improves prediction of downstream biological state $x_i(t+\tau)$ beyond controls:

$$I\big(u_{0:t}; x_i(t+\tau) \mid x_i(t)\big) > 0. \tag{14}$$

This criterion separates the informational hypothesis from a purely energetic hypothesis: the waveform must explain response variance after intensity, temperature, and nonspecific stress are accounted for.

## 7 A Minimal HOIF Re-entrainment Theorem

The full biological system is too complex to solve analytically. However, a simplified theorem clarifies what HOIF would require mathematically.

**Theorem: Local re-entrainment in a contractive HOIF system.** Let $S(t)$ evolve according to

$$\frac{dS}{dt} = F(S, A, t), \tag{15}$$

where $S \in \mathbb{R}^n$, and let $S_H$ be a locally stable healthy equilibrium under an input $A_H(t)$. Suppose that in a neighborhood $U$ of $S_H$ there exists a Lyapunov function $V(S) = \|S - S_H\|^2$ and constants $\lambda > 0$ and $\epsilon \geq 0$ such that

$$\nabla V(S) \cdot F(S, A, t) \leq -\lambda V(S) + \epsilon \tag{16}$$

for all $S \in U$ under a structured input $A(t)$. Then any trajectory that remains in $U$ satisfies

$$V(S(t)) \leq e^{-\lambda t} V(S(0)) + \frac{\epsilon}{\lambda}\big(1 - e^{-\lambda t}\big). \tag{17}$$

Thus, if $\epsilon$ is small relative to the initial drift, the state converges toward an $\epsilon/\lambda$ neighborhood of the healthy equilibrium.

**Proof.** The differential inequality $\dot{V} \leq -\lambda V + \epsilon$ implies, by Gronwall's inequality,

$$V(t) \leq e^{-\lambda t} V(0) + \int_0^t e^{-\lambda(t-s)} \epsilon \, ds. \tag{18}$$

Evaluating the integral gives

$$V(t) \leq e^{-\lambda t} V(0) + \frac{\epsilon}{\lambda}\big(1 - e^{-\lambda t}\big). \tag{19}$$

Therefore the distance to the healthy equilibrium decreases up to a residual error determined by model mismatch, noise, or insufficient coupling. □

**Interpretation.** The theorem does not prove that real cells can be re-entrained by sound. It states the mathematical condition that a HOIF-like control input would need to satisfy: the input must make the healthy region locally contractive relative to the disease-displaced state. This makes the biological question sharper: can any experimentally safe structured input produce a measurable negative drift derivative $\dot{D}(t) < 0$ beyond matched controls?

## 8 Stage 1: DNA Resonance Model

The first computational step estimates the natural frequency of a DNA base-pair opening coordinate using a simplified Peyrard–Bishop-style oscillator. The model treats the hydrogen-bond stretching coordinate as an effective nonlinear oscillator and estimates the small-amplitude natural angular frequency as

$$\omega_0 = \sqrt{\frac{2Da^2}{m}}, \tag{20}$$

where $D$ is the dissociation energy scale, $a$ is the inverse length scale of the Morse potential, and $m$ is an effective base-pair mass. The corresponding frequency is

$$f_0 = \frac{\omega_0}{2\pi}. \tag{21}$$

Using the parameters

$$\begin{aligned} m &= 4.982 \times 10^{-25} \text{ kg}, \\ D &= 0.033 \text{ eV}, \\ a &= 4.45 \times 10^{9} \text{ m}^{-1}, \end{aligned}$$

with $D$ converted to joules, the model produces a resonance near

$$f_0 \approx 1.03 \times 10^{11} \text{ Hz}, \tag{22}$$

or approximately 103 GHz. In the simulation, the response curve is narrow, meaning that the modeled structure responds most strongly near a specific frequency and weakly outside that band.

This result should not be interpreted as a claim that audible sound directly controls DNA. A frequency near 103 GHz lies far above the audible range. Instead, the result supports a more limited point: subcellular biological structures can be represented as frequency-selective systems, and some relevant modes may occur in microwave, millimeter-wave, or terahertz-adjacent regimes rather than ordinary sound.

## 9 Data-Driven Parameter Inference

The illustrative disease engine assigns drift values by hand. That is the weakest part of the current implementation. A rigorous HOIF model should replace assigned drift parameters with inferred parameters learned from biological measurements.

Let $Y$ denote observed biological data for a disease model. $Y$ may include RNA sequencing, single-cell RNA sequencing, ATAC-seq, chromatin imaging, proteomics, metabolomics, mitochondrial membrane potential, ATP production, cytokine levels, traction-force microscopy, cell stiffness, nuclear morphology, calcium signaling, or live-cell imaging. Instead of choosing a drift vector manually, HOIF should estimate

$$P(\delta \mid Y), \tag{23}$$

where $\delta = (\delta_m, \delta_c, \delta_{mit}, \delta_n, \delta_{chr}, \delta_{DNA})$ is the inferred multiscale drift vector.

A simple Bayesian version can be written as

$$P(\delta, \theta \mid Y) \propto P(Y \mid \delta, \theta) P(\delta) P(\theta), \tag{24}$$

where $\theta$ contains coupling strengths, layer weights, noise parameters, and response coefficients. The disease engine should then report posterior predictive quantities, not single fixed scores:

$$P\big(D(T) < D(0) \mid Y, A(t)\big). \tag{25}$$

- **RNA-seq / scRNA-seq:** transcriptomic distance to healthy cellular or regulatory state.
- **ATAC-seq / chromatin imaging:** chromatin accessibility, compaction, and nuclear-state deviation.
- **Proteomics:** pathway-level protein-state deviation across cellular and organelle layers.
- **Mitochondrial potential / ATP:** energetic-state deviation in the mitochondrial layer.
- **Cell stiffness / traction force:** mechanical-state deviation in membrane and cytoskeletal layers.
- **Cytokines / inflammatory markers:** immune-state deviation at tissue and cellular scales.
- **Live-cell imaging:** morphology, motility, and recovery trajectories across multiple layers.

This conversion changes the status of the model. The current simulation asks, "What follows if drift values are assumed?" A stronger HOIF implementation asks, "What drift values are implied by measured biology, and do they predict independent responses to structured input?"

## 10 Stage 2: Disease Testing Engine

The second step generalizes from a single molecular coordinate to a whole-cell disease-state model. Each disease is represented by a five-layer drift profile:

| **Layer** | **Healthy reference frequency** |
|---|---|
| Membrane | 200 |
| Cytoskeleton | 600 |
| Mitochondria | 1000 |
| Nucleus | 2000 |
| DNA-associated layer | 3000 |

These reference values are not asserted as measured biological constants. They are normalized modeling anchors used to compare layer-specific drift. A disease profile is then defined by percentage displacement from these healthy anchors. For example, a disease with a 20% mitochondrial drift is modeled as having a mitochondrial frequency of $1000 \times 1.20 = 1200$ in normalized units.

Each layer is weighted to reflect its contribution to the overall modeled disease state:

| **Layer** | **Weight** |
|---|---|
| Membrane | 0.15 |
| Cytoskeleton | 0.20 |
| Mitochondria | 0.25 |
| Nucleus | 0.25 |
| DNA-associated layer | 0.15 |

For each disease, recovery is simulated by applying a Kuramoto-style oscillator entrainment model. The diseased cellular oscillator begins at a shifted frequency and is coupled to a signal oscillator targeting the healthy frequency. The protocol includes three conceptual components:

- **Note:** a single healthy target frequency for each layer.
- **Chord:** simultaneous multi-layer targeting across membrane, cytoskeleton, mitochondria, nucleus, and DNA-associated dynamics.
- **Melody:** a time-varying sweep that begins near the diseased state and decays toward the healthy state, intended to improve capture into the target basin.

The modeled recovery percentage for a layer is calculated as

$$R = \max\left(0, \min\left(100, \left(1 - \frac{|f_{\text{final}} - f_{\text{healthy}}|}{|f_{\text{diseased}} - f_{\text{healthy}}|}\right) 100\right)\right). \tag{26}$$

The whole-disease recovery is the weighted sum of the five layer recoveries.

## 11 Biologically Interpretable Recovery Score

The illustrative recovery percentage in the disease engine is a mathematical distance score. It should not be read as clinical recovery. A data-driven implementation should define recovery using measurable biological endpoints. Let $z_k(t)$ be normalized biological features and $z_k^H$ their healthy reference distribution. A biologically interpretable recovery score can be defined as

$$R_{bio} = 1 - \frac{\sum_k \lambda_k \, d\big(z_k(T), z_k^H\big)}{\sum_k \lambda_k \, d\big(z_k(0), z_k^H\big)}, \tag{27}$$

where $d$ is a feature distance and $\lambda_k$ are endpoint weights learned or preregistered before analysis.

Examples of measurable recovery components include:

- restoration of mitochondrial ATP production or membrane potential;
- reduction of inflammatory cytokines relative to disease controls;
- normalization of nuclear morphology and chromatin accessibility;
- transcriptomic movement toward healthy reference samples;
- restoration of cytoskeletal organization or cell stiffness;
- reduced apoptosis, stress-response activation, or DNA-damage signaling.

A reported value such as "70% recovery" is meaningful only if the endpoints, weights, controls, and uncertainty intervals are stated. In future versions of HOIF, the preferred output should be a posterior distribution for $R_{bio}$, for example $R_{bio} = 0.70$ with a 95% credible interval of $[0.58, 0.79]$, rather than a single deterministic percentage.

## 12 Disease Database

The simulation tested 70 disease profiles across seven categories: cancer, neurological, genetic, autoimmune, metabolic, viral/infectious, and aging-related. Each disease was assigned a drift profile across the five modeled layers. This assigned database is intentionally broad and illustrative rather than clinically definitive. Its purpose is to demonstrate the structure of the engine, not to claim that the listed drift values are measured disease signatures. A publishable empirical version should replace this table with inferred drift distributions from public or newly collected biological datasets.

The disease categories were:

- **Cancer:** lung cancer, breast cancer, pancreatic cancer, leukemia, colon cancer, liver cancer, brain tumor, melanoma, prostate cancer, and ovarian cancer.

- **Neurological:** Alzheimer's disease, Parkinson's disease, ALS, multiple sclerosis, epilepsy, Huntington's disease, schizophrenia, bipolar disorder, depression, and autism spectrum conditions.
- **Genetic:** cystic fibrosis, sickle cell disease, muscular dystrophy, hemophilia, Down syndrome, Turner syndrome, Marfan syndrome, Tay–Sachs disease, PKU, and Fragile X syndrome.
- **Autoimmune:** lupus, rheumatoid arthritis, type 1 diabetes, Crohn's disease, psoriasis, celiac disease, Hashimoto's thyroiditis, Graves' disease, myasthenia gravis, and Sjogren's syndrome.
- **Metabolic:** type 2 diabetes, obesity, heart disease, atherosclerosis, hypertension, NAFLD, gout, osteoporosis, hypothyroidism, and hyperthyroidism.
- **Viral/infectious:** HIV, hepatitis B, hepatitis C, herpes, HPV, Ebola, tuberculosis, malaria, long COVID, and Lyme disease.
- **Aging-related:** aging, sarcopenia, macular degeneration, cataracts, dementia, arthritis, fibrosis, chronic inflammation, kidney disease, and heart failure.

## 13 Results

Across all 70 simulated diseases, the mean modeled recovery was 69.5%. This is an important result because it is not a universal near-perfect response. The model does not say that the protocol cures every disease. Instead, it produces a graded response that depends on the assumed structure of each disease.

The highest-scoring categories in the model were metabolic and autoimmune disease. These are conditions that, in this framework, are interpreted as involving substantial regulatory drift but less fixed structural corruption than cancer or inherited genetic disease. The lowest-scoring category was cancer, where the model assigned larger nuclear, mitochondrial, and DNA-associated drifts and therefore generated lower reversibility scores.

| **Category** | **Mean recovery** | **Interpretation** |
|---|---|---|
| Metabolic | 80.7% | Strongest modeled response |
| Autoimmune | 79.3% | Strong response, consistent with regulatory drift |
| Aging-related | 73.9% | Moderate-to-strong response |
| Neurological | 72.7% | Moderate response; complex layered tissue dynamics |
| Viral/infectious | 64.4% | Weaker response; pathogen-driven cellular rewriting |
| Genetic | 62.8% | Weaker response; encoded structural cause |
| Cancer | 53.0% | Weakest response; malignant autonomy and genomic disruption |

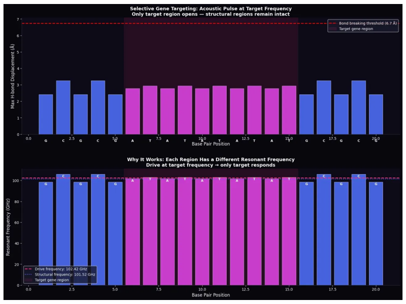
Several individual conditions were identified as high-priority candidates for future experimental modeling because they showed greater than 90% recovery in the simulation:

| **Condition** | **Modeled recovery** |
|---|---|
| Crohn's disease | 94.1% |
| Osteoporosis | 93.6% |
| Arthritis | 92.3% |

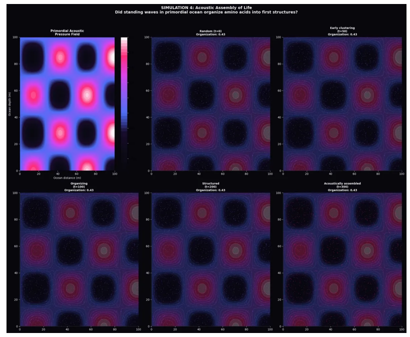
By contrast, cancer profiles showed substantially lower recovery, including severe simulated cases such as pancreatic cancer and brain tumor. This is useful primarily as a boundary condition for future tests. It suggests that acoustic entrainment, if experimentally supported, should be investigated cautiously as a possible adjunct variable rather than as a replacement for established cancer therapy.

## 14 Illustrative Simulation Image Trace

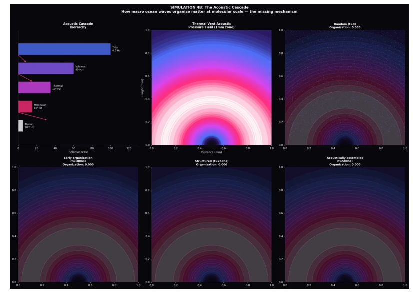
The uploaded experiment screenshots document the iterative development of the illustrative HOIF simulations, moving from molecular-scale resonance and acoustic organization toward cellular entrainment, structured temporal input, and disease-level ranking. These figures are included as a computational trace of the modeling process. They should be interpreted as simulation outputs and visual diagnostics, not as biological or clinical validation.



(a) Selective sequence-targeting output at the DNA-model scale.

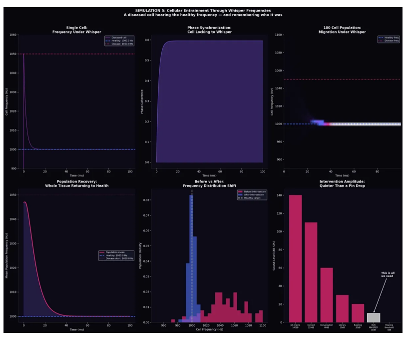


(b) Acoustic organization pattern across simulated spatial domains.



(c) Frequency-dependent organization and response maps.

Figure 4: Early-stage HOIF simulation panels. These panels document the transition from molecular and spatial organization models toward the broader hypothesis that structured oscillatory inputs can be treated as controllable variables.



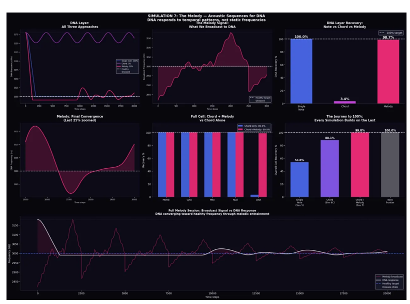
(a) Intermediate acoustic cascade diagnostics.

(b) Multi-layer chord-style entrainment output.

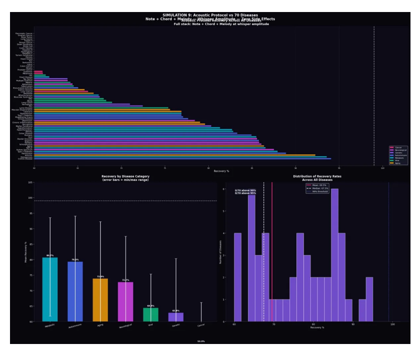
(c) Fixed full-chord entrainment comparison.

Figure 5: Middle-stage simulation panels. These panels show the progression from single-layer response assumptions toward simultaneous multi-layer input, motivating the note–chord–melody structure used later in the disease engine.

(a) Kuramoto-style full-chord oscillator output.



(b) Melody-style temporal sequence output for DNA-layer response.



(c) Disease-engine category and distribution summary.

Figure 6: Late-stage simulation panels. The final panels summarize the shift from static frequencies to structured temporal input and then to disease-category ranking. The disease panel is included to document the model-generated hierarchy, not to claim clinical efficacy.

# 15 Interpretation

The main output is a disease-response map. In this model, oscillatory-state intervention is most worth testing when disease is represented as dynamic drift rather than irreversible blueprint damage. This leads to a three-tier therapeutic interpretation.

## 15.1 Tier 1: Acoustic Protocol as Primary Intervention

Metabolic, autoimmune, and selected aging-related diseases may be the most plausible first targets. These conditions often involve chronic dysregulation, inflammation, altered cellular signaling, mitochondrial stress, tissue remodeling, or maladaptive homeostasis. In the present framework, such states are closer to healthy attractors and therefore easier to entrain.

The most practical laboratory starting points are not necessarily the most dramatic diseases. They are the diseases with high modeled recovery, large patient burden, and measurable cellular endpoints. Crohn's disease, osteoporosis, and arthritis are therefore proposed as first-pass experimental targets.

## 15.2 Tier 2: Acoustic Protocol as Supportive Intervention

Neurological and viral/infectious conditions showed intermediate modeled recovery. These categories may require combination approaches. In neurological disease, the challenge is tissue complexity, cell-type specificity, blood–brain barrier delivery, and network-level behavior. In viral and infectious disease, the challenge is that the disease process may involve an external replicating agent or persistent intracellular program. Here, acoustic stimulation may be better viewed as a way to restore cellular environment, reduce stress, or support conventional treatment rather than replace it.

## 15.3 Tier 3: Acoustic Protocol as Preparation or Sensitization

Cancer and inherited genetic diseases showed the lowest modeled recovery. This is consistent with the framework because these diseases involve deeper alteration of cellular identity or genetic architecture. Sound or oscillatory stimulation alone should not be expected to repair a mutation, remove malignant clones, or correct inherited DNA sequence defects.

However, this does not make the approach irrelevant. A more realistic cancer or genetic-disease hypothesis is that frequency-based stimulation could prepare tissue for another intervention. For cancer, it may alter membrane permeability, cytoskeletal tension, mitochondrial state, or drug sensitivity. For genetic disease, it may support delivery or expression of gene-targeted therapies. In this tier, the acoustic protocol is not the cure; it is a possible amplifier of precision medicine.

## 16 Information-Theoretic Framing

HOIF is broader than acoustics. Acoustic, ultrasonic, electromagnetic, or mechanical inputs are possible carriers; the central claim concerns information. The experimentally relevant question is whether input structure reduces uncertainty about future biological state. In information-theoretic terms, a structured input is meaningful only if

$$I(A_{0:T}; S(T) \mid E, \bar{f}, T_{exp}, \Delta T) > 0, \tag{28}$$

where $E$ is delivered energy, $\bar{f}$ is average frequency, $T_{exp}$ is exposure duration, and $\Delta T$ is temperature change. This conditional formulation is important: it prevents the framework from confusing information with dose.

A negative result is also informative. If $I(A_{0:T}; S(T) \mid E, \bar{f}, T_{exp}, \Delta T) = 0$ across carefully controlled experiments, then waveform organization is not carrying detectable biological information in that system. HOIF is therefore falsifiable: it succeeds only where temporal organization improves prediction beyond standard physical exposure variables.

## 17 Key Testable Prediction

The most useful single prediction of this framework is waveform specificity:

> If hierarchical mechanical information processing is a useful model, then waveform structure—not only total energy, peak intensity, or carrier frequency—should explain reproducible differences in biological response under controlled nonthermal conditions.

This prediction is more specific than the broad claim that cells respond to mechanical stimulation. It can be tested by asking whether two signals matched for exposure duration, intensity, temperature change, and average frequency still produce measurably different outcomes because their temporal structure differs. A minimal experiment would compare four groups: sham exposure, single-frequency exposure, random-frequency exposure, and structured note–chord–melody exposure, all matched for delivered energy. The framework would be supported only if the structured waveform produced a response pattern that could not be explained by heating, dose, or nonspecific mechanical stress alone.

## 18 Baseline Models and Empirical Validation

HOIF should be benchmarked against simpler alternatives. A framework is useful only if it predicts something that baseline models miss. Candidate baselines include:

- **Energy-only model:** response depends only on delivered energy and exposure duration.

- **Frequency-only model:** response depends on carrier frequency but not waveform organization.
- **Linear control model:** biological state changes linearly with input amplitude.
- **Random stimulation model:** structured input performs no better than random temporal organization.
- **Network diffusion model:** response propagates passively through a biological network without waveform-specific control.
- **Generic stress model:** observed changes are explained by heat-shock, inflammatory, apoptosis, or DNA-damage pathways.

The proposed framework should be evaluated against at least one published mechanobiology or ultrasound-bioeffects experiment before being used to predict new experiments. The ideal validation procedure is:

1. select a published dataset with known mechanical or oscillatory perturbation and measured biological outputs;
2. infer HOIF drift and coupling parameters using only the training subset;
3. predict held-out biological responses;
4. compare predictive accuracy against the baseline models above;
5. report whether waveform structure improves prediction after controlling for energy, duration, temperature, and stress markers.

This requirement prevents HOIF from becoming only a speculative simulation. If it cannot reproduce or explain existing experiments better than simpler models, it should not be treated as a useful scientific framework.

## 19 Proposed Laboratory Roadmap

The next step is not clinical use. The next step is controlled laboratory testing. A practical validation pathway would proceed as follows:

1. **Select one high-response disease model.** Begin with a cell culture model related to Crohn's disease, osteoporosis, or arthritis rather than cancer.
2. **Define measurable endpoints.** Use inflammatory markers, mitochondrial function, cell viability, calcium signaling, cytoskeletal organization, gene-expression panels, and morphology.
3. **Map baseline signatures.** Measure untreated healthy-like and disease-like cellular states using mechanical, electrical, optical, and molecular assays.

4. **Apply frequency sweeps safely.** Test low-intensity, nonthermal exposure across candidate frequencies while monitoring temperature, cavitation risk, and nonspecific stress, following safety concerns established in therapeutic ultrasound and sonoporation research [5, 6].

5. **Compare against controls.** Include sham exposure, random-frequency exposure, known drug treatment, and combined treatment groups.

6. **Test specificity.** A real effect should show frequency dependence, dose dependence, reproducibility, and biological specificity rather than nonspecific heating or damage.

Only if repeatable cell-culture results are obtained should the framework move to organoids, tissue slices, animal models, or human feasibility studies.

# 20 Uncertainty Quantification and Monte Carlo Robustness

A HOIF disease engine should report uncertainty, not only point estimates. For each disease or experimental condition, parameters should be sampled from distributions:

$$\theta^{(j)} \sim P(\theta \mid Y), \qquad j = 1, \dots, N, \tag{29}$$

where $N$ should be large enough for stable intervals, for example $N = 10{,}000$ in a computational sensitivity study. Each sample produces a recovery or state-shift estimate $R^{(j)}$. The reported output should include

$$\hat{R} = \text{median}\{R^{(j)}\}, \qquad CI_{95} = \left[q_{0.025}(R), q_{0.975}(R)\right]. \tag{30}$$

Ranking claims should also be probabilistic. Instead of writing "Crohn's disease is top-ranked," a robust version would report

$$P\big(\,\text{rank(Crohn)} \le 5 \mid Y\big) = 0.91, \tag{31}$$

if Crohn's disease remains in the top five candidates in 91% of Monte Carlo simulations. This kind of statement is much stronger than a single deterministic table because it exposes model fragility.

# 21 Sensitivity Analysis Requirement

Because the current disease engine depends on assigned drift values, layer weights, coupling strength, and protocol timing, HOIF should not be evaluated from a single parameter setting. A publishable version of the model should include a sensitivity analysis in which the following quantities are perturbed by at least ±30%:

- layer weights $w_i$;
- healthy reference frequencies $f_i$;
- disease drift magnitudes $\delta_i$;

- coupling strength $K$;
- simulation duration and sweep timing;
- stochastic noise level $\eta(t)$.

The key robustness question is not whether the exact recovery percentages remain unchanged. They should not be expected to. The key question is whether the category-level ranking remains qualitatively similar across perturbations. A minimal robustness criterion is

$$\rho\big(R_{base}, R_{perturbed}\big) > 0.7, \tag{32}$$

where $\rho$ is the rank correlation between baseline and perturbed category-level recovery scores. If the hierarchy collapses under modest perturbation, the disease-engine result should be treated as a parameter artifact. If it remains stable, the model becomes more useful as a prioritization tool for experiments.

- **Layer weights:** perturb by $\pm 30\%$ and renormalize; pass if category rank correlation remains $> 0.7$.
- **Reference frequencies:** perturb by $\pm 30\%$; pass if rankings do not collapse to random ordering.
- **Disease drifts:** perturb by $\pm 30\%$; pass if top candidate classes remain identifiable.
- **Coupling strength:** perturb by $\pm 30\%$; pass if the response hierarchy remains sufficiently monotonic to rank.
- **Noise:** test low, medium, and high noise; pass if prediction intervals remain interpretable.

# 22 Alternative Interpretations

A strong test of the framework must distinguish the proposed information-bearing effect from simpler explanations.

**General mechanotransduction.** Any observed response may reflect ordinary mechanotransduction rather than a distinct acoustic information layer. This possibility should be tested by comparing structured waveforms against energy-matched unstructured mechanical stimulation.

**Thermal effects.** Apparent frequency-specific effects may result from small temperature changes. Experiments must include real-time thermal monitoring and temperature-matched controls.

**Cavitation and membrane damage.** Ultrasound can perturb membranes through cavitation or sonoporation. If the biological response is driven by membrane disruption, it should correlate with cavitation markers or permeability assays rather than waveform information.

**Stress-response gene expression.** Changes in transcription may reflect generic cellular stress rather than beneficial state correction. Gene-expression analysis should therefore distinguish inflammatory, heat-shock, DNA-damage, and apoptosis pathways from disease-relevant normalization.

**Model-parameter artifact.** The disease hierarchy may arise from the assigned drift values and layer weights rather than from a robust biological principle. Sensitivity analysis should vary weights, drift magnitudes, coupling strength, noise, and protocol timing to test whether the category pattern persists.

**Placebo or observer effects in later studies.** If the framework advances beyond cell culture, blinded protocols and preregistered endpoints will be essential. The present paper makes no patient-level claim.

## 23 Falsification Criteria

HOIF should be rejected, or at least substantially revised, if any of the following occur repeatedly under well-controlled conditions:

1. waveform structure never outperforms energy-matched and frequency-matched controls;
2. apparent information-specific effects disappear under blinded analysis;
3. inferred drift parameters fail to predict independent datasets;
4. the model cannot reproduce established mechanobiology experiments better than simpler baselines;
5. biological responses are fully explained by thermal effects, cavitation, membrane damage, or generic stress pathways;
6. Monte Carlo sensitivity analysis shows that rankings collapse under modest parameter perturbation;
7. posterior predictive intervals are so wide that the framework cannot generate useful experimental priorities.

These criteria are intentionally strict. The goal is not to protect HOIF from failure; the goal is to make failure informative.

## 24 Limitations

This study has major limitations.

First, the disease drift values in the present implementation are hypothetical modeling inputs. This is the main limitation. They must be replaced by drift parameters inferred from real biological measurements before disease-specific conclusions can be drawn.

Second, the healthy layer frequencies are normalized anchors rather than established universal biological constants. They allow comparison within the model but should not be treated as clinical frequencies.

Third, the Kuramoto model is a simplified entrainment model. Real cells are nonlinear, heterogeneous, adaptive, noisy, spatially structured, and biochemically regulated. Entrainment of a mathematical oscillator does not guarantee therapeutic reprogramming of a living cell.

Fourth, acoustic and electromagnetic delivery are unresolved. Frequencies relevant to DNA-scale motion may not propagate through tissue in the same way as audible sound or ultrasound. Biological safety, absorption, heating, scattering, and targeting must be determined experimentally.

Fifth, recovery percentage in this paper is a model-defined mathematical score, not a clinical endpoint. A 90% modeled recovery does not mean a patient would improve by 90%.

These limitations are not minor. They define the boundary between a testable computational framework and any future validated medical technology.

## 25 Conclusion

This paper presents a hypothesis-generating disease testing engine for evaluating whether acoustic or oscillatory inputs are worth testing as modulators of cellular state. The framework begins with the idea that biological structures have frequency-selective dynamics and that disease can be modeled as drift away from healthy resonant behavior. A DNA base-pair oscillator model produced a narrow theoretical resonance near 103 GHz under the selected parameters. A broader Kuramoto-style disease engine tested 70 disease profiles and produced a mean modeled recovery of 69.5%.

The most important result is not evidence that the protocol works clinically. It is not clinical evidence. The useful result is the hierarchy of testable predictions. Metabolic and autoimmune diseases show higher modeled response; cancer and genetic diseases show lower modeled response. This pattern is biologically plausible enough to motivate targeted testing because reversible regulatory drift may be more experimentally tractable than mutation-driven or malignancy-driven disease.

The framework therefore suggests a disciplined research strategy: do not begin with claims of universal cure. Begin with controlled cell-culture experiments in high-response, high-burden, currently undertreated conditions such as Crohn's disease, osteoporosis, and arthritis. If reproducible frequency-specific effects are observed there, the model can be refined, expanded, and tested in more complex disease systems.

The cell may not literally "hear" sound, but it can respond to rhythm, force, vibration, and mechanical context. The question worth testing is whether structured oscillatory inputs contain enough controllable information to shift biological state in reproducible, specific, and safe ways. This paper offers a computational rationale for asking that question experimentally.

## 26 Computational Reproducibility Requirements

A publishable HOIF implementation should be distributed with the following materials:

- source code for all simulations;
- disease or cell-state parameter files;
- scripts for Bayesian parameter inference;
- Monte Carlo sensitivity-analysis scripts;
- baseline-model comparison scripts;
- random seeds and environment specifications;
- raw and processed datasets or links to public repositories;
- a reproducible notebook that regenerates every table and figure.

Without these materials, the disease engine should be described as illustrative. With them, independent researchers can test whether HOIF survives changes in assumptions, datasets, and modeling choices.

## Acknowledgment of Scientific Status

This manuscript is a theoretical and computational proposal. It should be read as a framework for generating experimentally testable hypotheses, not as evidence that acoustic stimulation treats, reverses, or cures any disease. Any future biological testing would require institutional review, appropriate biosafety procedures, calibrated exposure equipment, thermal and mechanical dosimetry, and independent replication.

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